AMERICAN UNIVERSITY OF BEIRUT

Faculty of Arts and Sciences
Department of Computer Science
CMPS 261 - Machine Learning
Layout preview - 60 min

Total: 10 marks

Name: _____ Student ID: _____

This exam has 2 questions, worth 10 marks in total. You have 60 minutes. For multiple-choice questions, choose one answer.

1) Explain the observation and justify your answer. *[5 marks]*

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2) A logistic classifier assigns the following disease probabilities. *[5 marks]*

Probability p	Actual disease present	Actual disease absent
0.82	3	0
0.56	1	5
0.34	1	6
0.12	0	4

Predict disease present when $p \geq t$. Maximize precision subject to recall of at least 80%.

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$J = \frac{1}{4}\sum_{i=1}^{4} - y_i \ln p_i$$

a) Select $t = 0.50$.

b) Select $t = 0.70$.